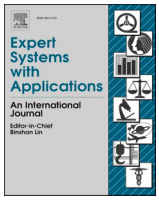
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An intelligent BIM-enabled digital twin framework for real-time structural health monitoring using wireless IoT sensing, digital signal processing, and structural analysis

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ABSTRACT

Structural health monitoring (SHM) of civil infrastructure is critically important due to its direct influence on public safety and economic activities. Exiting Building Information Modeling (BIM)-based SHM systems often use offline data processing techniques to analyze and visualize structural health data. Despite some of them adopting Internet of Things (IoT) to enable real-time sensor data collection, sensor data quality still remains uncertain due to a lack of sensor signal preprocessing in those existing systems. Additionally, the IoT-based SHM systems often disregard structural analysis domain knowledge, which is important for accurate and precise SHM. Therefore, there is still a need to improve existing systems and practices by enabling more efficient and reliable data collection and processing as well as providing more representative SHM data visualization in BIM. As such, this paper proposes an intelligent BIM-enabled digital twin (DT) framework that integrates wireless IoT sensing and communication, digital signal processing (DSP), and structural analysis domain knowledge. The proposed system (1) leverages IoT sensing and wireless communication to enable autonomous and real-time SHM sensor data collection and transmission, (2) applies and compares multiple DSP techniques to preprocess the sensor data/signals, and (3) innovatively embraces structural analysis expertise into structural behavior visualization in BIM by performing sensor data interpolation for enabling the visualization of structural behaviors at different locations of a structural element/component. The proposed BIM-enabled DT framework was demonstrated and tested for monitoring and visualizing the structural deformations of critical structural components using a prototyped structural frame subject to bending forces. The developed framework could be used and extended for any structural elements (such as beams, columns, trusses, slabs, arches, bracings, walls, footings, foundations, and girders, among others) and could be applied to any kind of structure. Experimental results showed that the proposed framework could effectively monitor and intuitively visualize the structural deformations under different load configurations with a high DT updating frequency of 5 Hz. The innovation of this study is reflected by integrating structural analysis expertise with IoT-enabling sensing data analytics in order to improve the representativeness of real-time structural behavior visualization in BIM and to advance the DT-based SHM systems in a faster and more adaptive direction. Ultimately, this paper contributes to the body of knowledge by developing a generic and easily extendable BIM-enabled DT framework for SHM with high sensor data quality and improved visualization to advance the existing practices of BIM-based SHM for civil infrastructure asset management.

1. Introduction

Civil infrastructure encompasses a wide range of physical structures and facilities (such as buildings and bridges) that play a crucial role in the development and functioning of societies, such as providing the foundation for economic growth, public safety, and overall well-being (Hu et al., 2024). Thus, the health condition of civil infrastructure is critically important to society due to its direct impact on public safety, economic activities, quality of life, resilience to natural disasters, and even public trust (Osei-Kyei et al., 2021). For example, a deteriorating

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bridge in poor condition has high risks of collapse when there is no timely and insufficient maintenance (Assaad and El-adaway, 2020; Hu and Liu, 2023; Hu and Assaad 2024a). As a result, a collapsed bridge will lead to disruptions in transportation, injuries and potential loss of life, delays in the delivery of merchandises and emergency response, challenges due to disruption to transportation network, cascading damages to other surrounding infrastructure such as pavements, and erosion of public confidence (FHWA, 2023). Therefore, it is significantly necessary to dynamically monitor the conditions of the civil infrastructure or its critical components by providing real-time data on the structural integrity, performance, and safety.

Capitalizing on the advancement of the Internet of Things (IoT) technology, plenty of IoT-based structural health monitoring (SHM) systems have been developed for a variety of applications in the civil infrastructure sector (Tong et al., 2019; Peng et al., 2023; Hu and Assaad, 2023). More specifically, IoT-based SHM uses an interconnected network of sensors (e.g., strain gauges) and other devices such as 4G modem embedded in the infrastructure. It collects data on many parameters such as strain and vibration frequency and then transmits the data to a centralized platform using different communication protocols (e.g., Bluetooth, LoRaWAN, and Zigbee) (Gao et al., 2018; Assaad et al., 2023). Afterwards, the data is processed, analyzed, and made accessible to infrastructure asset owners or other stakeholders for decision-making purposes.

Empowered by more advanced communication technology and stronger computational capability of machines, digital twin-enabled IoT-based SHM systems have been developed recently for real-time data transmission and processing. A digital twin (DT) is a virtual replica of a physical object that mimics its real-world counterpart by incorporating data from various sources such as sensors, simulations, and historical information (Botín-Sanabria et al., 2022).

Building information modeling (BIM), on the other hand, provides many added benefits and advantages. In fact, BIM has been widely used to supplement IoT-based SHM by (1) creating a digital model of a structure with incorporated geometric information, materials, and components, (2) providing a user-friendly visualization environment to display the structural behavior and performance, and (3) offering high customizability to multiple items such as digital model's property, level of detail, and view interface (O'Shea et al., 2020; Gragnaniello et al., 2024). Therefore, it is beneficial to integrate IoT-based SHM with BIM because the research on BIM-IoT integration is still considered to be in the initial phases; hence, requiring substantial efforts to gain a deeper understanding of the current situation (Tang et al., 2019).

In addition, most of the existing studies on IoT-based SHM barely addressed the sensor data quality issue that could be caused by noises or missing values before transmitting the data for subsequent analysis (Sakr and Sadhu, 2023). Lack of adequate SHM data pre-processing unavoidably reduces the reliability of IoT-based SHM systems due to uncertain sensor data quality.

While some studies have attempted to integrate BIM and IoT for SHM (Scianna et al., 2022), they commonly have a lack of demonstrating the interoperability of their BIM models and do not justify the rationale for guiding sensor installation on the structure or structural components either.

Therefore, there is still a need to enhance the existing practices of BIM-enabled and IoT-based SHM by handling the sensor data quality issue, enabling BIM model interoperability, as well as explaining the logic followed for guiding sensor installation. To address this need, the goal of this paper is to develop an intelligent BIM-enabled digital twin framework that embraces IoT sensors, wireless communication, digital signal processing, and structural analysis domain knowledge for autonomously sensing and visualizing the structural deformation of the critical components of a structure in a real-time manner.

More specifically, an intelligent BIM-enabled digital twin framework represents a sophisticated integration of BIM, digital twin technology, and intelligent systems (Hu and Assaad 2024b). BIM provides a digital representation of physical and functional characteristics (e.g., geometry, and material properties) of an infrastructure and/or its critical components. This digital representation will be used as the base to develop a digital twin model of the physical structure. A digital twin model is a virtual representation of the physical infrastructure, which works as a dynamic digital counterpart that mirrors the behaviors, status, and performance of its real-world counterpart in real-time. The real-time correspondence between the virtual and physical counterparts are enabled by intelligent systems with high-speed communication capability. An intelligent system refers to technologies such as artificial intelligence, machine learning, and data analytics allowing machines to perform tasks that typically require human intelligence. Such a system can collect and analyze vast amounts of data from IoT sensing networks or other sources to identify patterns and perform predictions with minimal human intervention. Therefore, BIM-enabled digital twins for SHM hold great significance because of its capability of allowing (1) comprehensive data integration to enable a holistic understanding of the structure's behaviors, performance, and condition (Sharma et al., 2022), (2) early detection of structural issues for supporting timely interventions to prevent structural failures or accidents, and (3) implementation of preventative maintenance strategies to extend the lifespan of the structure, minimize downtime, and reduce maintenance costs (Errandonea et al., 2020).

The contributions of this study are summarized as follows:

- Proposing a BIM-enabled digital twin framework with much higher updating frequency to advance the development of real-time SHM systems toward faster and more adaptive direction
- Incorporating digital twins with BIM interoperability to allow seamless real-time SHM data exchange for supporting structural analysis, smart sensing, and structural behavior visualization
- Innovatively integrating the rich structural analysis domain knowledge with sensing data analytics to unleash the potential of existing BIM software for more intuitive structural behavior representation and visualization

2. Literature review

The existing body of knowledge on SHM is huge and includes various studies with multiple applications and methods. Since it is not feasible or possible to review all existing studies in the broader SHM field, this section focuses on relevant studies that have considered the use of BIM for SHM-related applications and approaches. This particular focus helps in offering better insights on the existing relevant literature and hence better identifying and highlighting the knowledge gaps that this paper specifically aims to address.

2.1. BIM-based SHM: General work mechanism and applications

BIM functions as a comprehensive and digital representation of the physical and operational attributes of a structure. BIM emerges as an indispensable tool to optimize and document the performance of a structure across its lifespan. Based on its strong capability of information modeling and visualization, BIM enables a move toward intelligent management-related practices of structures with the use of smart sensing (Kensek, 2014). Using facility management (FM) as an example, previous research endeavors have sought to merge BIM with a variety of sensing technologies and devices across multiple domains within building automation (Abideen et al., 2022; Altohami et al., 2021). These studies focus on energy management in residential buildings, and employing sensors for measuring temperature, humidity, and energy consumption (Abdelalim et al., 2017; Chang et al., 2018; Costa et al., 2013; Ozturk et al., 2012; Xie et al., 2020). Meanwhile, some other studies particularly concentrated on structural monitoring and assessment, either within the framework of operations and maintenance (O&M) or under the umbrella of SHM (Panah & Kioumarsi, 2021).

The integration of BIM and SHM aims to incorporate SHM data into the information stored and presented within the BIM model for updating the model's attributes or properties to assist in evaluating the structure's performance and condition. Numerous conceptual frameworks have previously focused on fusing sensor data into BIM elements by generating customized BIM families and establishing links with sensor data for interpretation and storage purposes (Banfi et al., 2017; Davila Delgado et al., 2017; O'Shea & Murphy, 2020; Theiler et al., 2017; Valinejadshoubi et al., 2017; Zhang et al., 2015). For instance, identifying structural anomalies and damages is also a significant application of BIM-SHM integration (Panah & Kioumarsi, 2021). In summary, researchers commonly utilize BIM-based SHM for monitoring and assessing changes in structural properties based on the data obtained from sensors (Delgado et al., 2018; Hess et al., 2018; Wang et al., 2017).

2.2. BIM-based SHM: Structural behavior visualization

Previous studies also revealed that analyzing sensor data alone does not substantially contribute to understanding the facility's condition, as the data makes far more sense when analyzed within a relevant "context" (Liu and Akinci, 2009). This context can be supplied by spatial information derived from BIM models, which enables a visualized and unified approach to data analysis.

In order to enhance the visual aspects of the monitoring process, researchers have experimented with various software and visual programming environments. Efforts have been made to link sensor data with BIM for plenty of applications such as building maintenance management to attain comprehensive control over project and structural monitoring (Kensek, 2014). Kazado et al. (2019) aimed to visualize and analyze real-time building information by developing a user-friendly add-in program that streams sensor data into the BIM process using Navisworks software. Chang et al. (2018) introduced a platform focusing on translating sensor data into visual representations by employing a color-coded visualization in a BIM model to present multiple sensor data sets visually. Valinejadshoubi et al. (2019) developed a framework to transfer the real-world sensor data into virtual one and then detect structural damages and visualize them in BIM through color coding. Delgado et al. (2018) integrated fiber optic sensor data into BIM, visualizing structural deformations through color coding and dynamic charts. Additionally, Boddupalli et al. (2019) proposed a framework for digitally representing the collected large volumes of vibration sensor data from bridges, analyzing the sensor data for bridge SHM, and linking results in a tabular form to sensor parameters defined in the BIM environment.

2.3. Knowledge gap

Despite the importance of existing studies about BIM-based SHM, three primary knowledge gaps still exist.

First, majority of existing BIM-based SHM systems used offline processing techniques to analyze the heterogenous data (e.g., vibration sensor data) and visualize the structural behaviors in BIM software for supporting maintenance decision making. An offline processing technique refers to the handling or analysis of data without real-time interaction, which often results in a significant delay between data collection and data analysis (Ed-daoudy and Kaalmi, 2019). However, early warning of the health conditions of a civil infrastructure often requires more efficient data processing to evaluate structural safety and reliability in time. Nowadays, real-time/online processing has become more and more popular for SHM due to more affordable sensing technologies and faster communication capability, which provides a great opportunity to develop DT-based intelligent SHM systems. In fact, a DT-based SHM system relies on wired or wireless IoT connectivity, real-time data processing, and graphical representation of physical civil structures. Despite that many studies proposing different frameworks to integrate sensor data into BIM for structural behavior visualization in the context of DT, they commonly used historical sensor data collected without IoT connectivity. Meanwhile, based on IoT, a few studies claimed real-time BIM-based SHM, but they suffer from either high latency due to long response time or low transparency caused by unreported response time. For instance, Wang et al. (2022) developed a real-time BIM-based SHM and early warning system for timber heritage structures with a response frequency at a daily base. In fact, a response frequency of one day is associated with high latency, resulting in delayed decision making, low capability of reacting to rapid events (e.g., earthquakes), and low system flexibility (Pons et al., 2023). Another study by Fawad et al. (2023) proposed a real-time bridge SHM system based on IoT, BIM, finite element analysis, and field loading test. However, there is missing information of how fast that system can collect and process the data from the field using IoT devices. It further hinders the transparency and reliability of the BIM-based bridge SHM system. Thus, there is still a need to develop a BIM-enabled DT system for SHM using smart sensing and wireless communication for high-speed real-time data transmission, processing, and visualization in support of early warning of poor structural conditions, which is the focus of this paper.

Second, little-to-no studies have emphasized sensor data quality issues such as noises and missing values for IoT-based SHM. As a result, the reliability and trustworthiness of those works would remain unsure due to a lack of sensor data/signal preprocessing for addressing the data quality issues. Therefore, this paper applies and compares different digital signal processing techniques to preprocess the sensor signals in order to increase the trustworthiness of IoT-based SHM systems based on sensor data.

Third, most of the existing BIM-based SHM systems commonly apply color coding methods to visualize the structural behaviors (e.g., deformation) using one sensor reading to colorize an entire element of a structure. In other words, when there is a sensor installed on one structural element, the entire element will be visualized using a uniform color based on the sensor reading. While such visualization method might reflect the overall performance level of the element in a global way, it has a lack of differentiating the structural responses at different locations of the structural element. Therefore, it may suffer from low representativeness when there is a need to visualize the heterogenous behaviors of a structural element. Thus, this paper innovatively proposes the use of structural analysis domain knowledge to perform sensor data interpolation for enabling the visualization of different structural behaviors in BIM using measurements from one single sensor. In this way, it would not only increase the SHM visualization representativeness in BIM but also have great potential to improve the cost-effectiveness of the IoT-based SHM systems due to less sensors being required.

3. Methodology

3.1. Overview

Fig. 1 presents the methodology followed by this paper, which contains three modules, including (1) digital asset development, (2) physical asset development, data sensing, preprocessing, validation, and transmission, and (3) BIM-enabled SHM digital twin creation and visualization. Details about each module are provided in the following subsections.

Table 1 summarizes the major tools used in this study to implement the proposed BIM-enabled digital twin framework for SHM.

As shown in Table 1, these tools are comprised of both software and hardware, as detailed in the next subsections.

3.2. Module 1: Digital asset development

Digital asset development aims to generate a 3D virtual structural model of a physical structure/asset and then conduct structural analysis on the model for identifying the asset's critical components and the

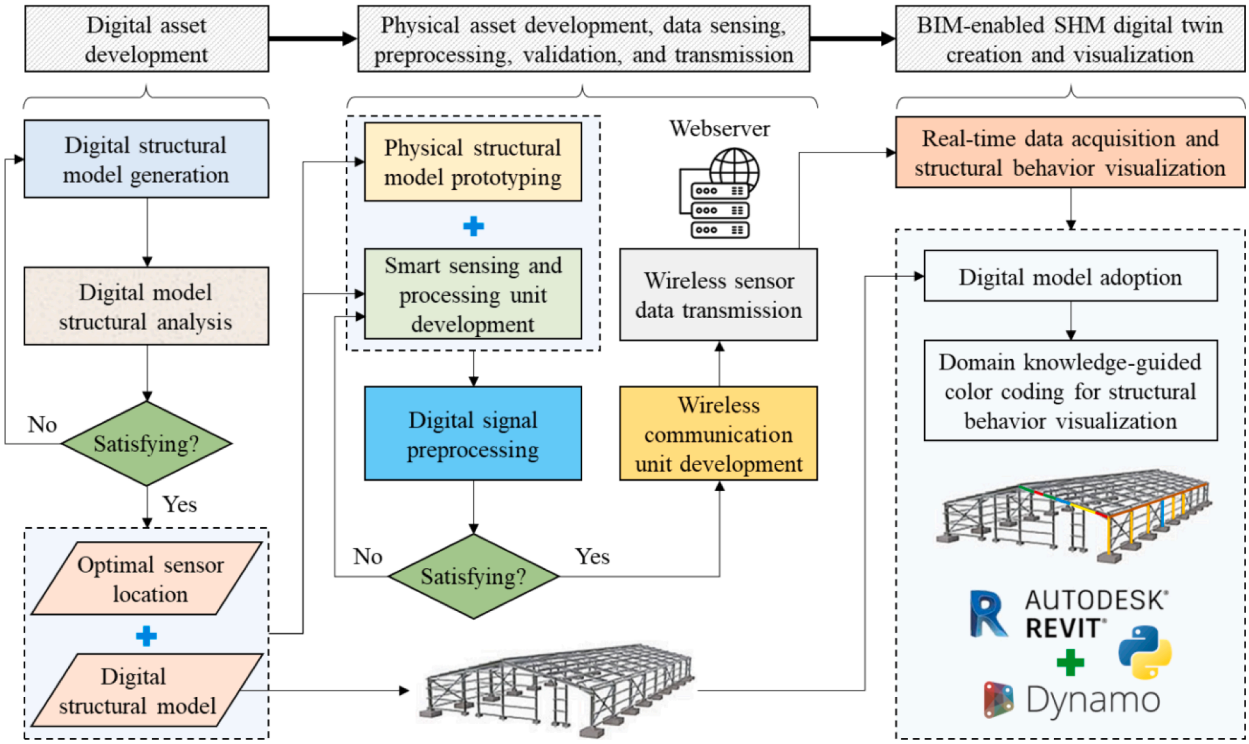


Fig. 1. Methodology for developing the BIM-enabled digital twin for SHM.

potential damage locations under both external and internal forces. More specifically, it is a process of 3D modeling that involves the creation of virtual representations of physical structures or components using computer-aided design software. These digital assets serve as accurate and detailed models that capture the geometry, material properties, and other relevant characteristics of the structure. Through meticulous modeling, engineers can replicate real-world structures in a virtual environment, enabling comprehensive analysis and simulation of their behavior under various conditions.

The Robot Structural Analysis Professional 2023 (a structural analysis computer software tool) was used to develop the digital assess and perform structural analysis; hence, generating a range of results that provide insights into the behaviors of the structure under different loading conditions. Fig. 2 shows the created digital model for a building structural frame prototype, which is comprised of 12 elements made of 6063-T52 aluminum. The 12 structural elements include four strip footings (3.5 mm × 31.8 mm × 350.0 mm) on the base, four columns (6.2 mm × 25.4 mm × 300.0 mm), two type-A beams (1.6 mm × 25.4 mm × 300.0 mm), and two type-B beams (3.2 mm × 12.7 mm × 300.0 mm). The beam and column elements are numbered separately using green and blue colors, respectively, as shown in Fig. 2.

This study specifically focuses on two types of structural deformation (i.e., deflection and strain) due to their popularity in performance-based design and the assessment of structural safety (Wei et al., 2021). To ensure the validity of the structural analysis results, self-checking-based hand calculations are implemented to check the potential errors in the computer results (Hanson, 2022). The self-checking-based hand calculations refer to the use of standard formulations to perform manual calculation of the responses (e.g., stress, and stain) at a given location of a structural element of interest under pre-defined load conditions. Satisfactory structural analysis results occur when computer-generated outcomes closely match hand calculations. In cases where there is a discrepancy, revising the digital structural model becomes necessary, followed by repeating the structural analysis and conducting error checks.

As a result, the digital model development module in this paper generates two major outputs to support physical asset development, sensing unit development, and BIM-enabled digital twin creation and visualization. One output contains the bending moment diagrams for all components of the structure to guide the installation of SHM sensors (e.g., strain gauges) at the places where the maximum bending moment(s) occur. The other output is an interoperable digital model that can be used in different BIM software tools seamlessly. The interoperability refers to the characteristic of the digital model to work with different BIM software tools. More specifically, the digital model of the structural frame was developed using the BIM software named Robot Structural Analysis Professional 2023. This software was also used to perform structural analysis on the created digital model. Meanwhile, due to the interoperability of the digital model, it was later imported into another BIM software (Autodesk Revit 2024) for structural behavior visualization based on streaming sensor data and color coding. The interoperability of the digital structural model enabled by Industry Foundation Classes (IFCs) allows for an efficient data exchange between software without information loss, elimination of reworks and errors, and a smooth transition from construction to facility management (Ghaffarianhoseini et al., 2017; Sibenik and Kovacic, 2020).

3.3. *Module 2: Physical asset development and data sensing*

This section details the steps followed for developing the prototype for a physical structure as well as for sensing, preprocessing, validating, and transmitting the SHM data.

3.3.1. *Developing a physical prototype*

Physical asset development generally involves constructing either an experimental prototype or a full-scale real-world structure, both derived from the digital model (Kim, 2019). Inspired by recent studies that have used experimental prototypes for BIM-based SHM (Sakr and Sadhu, 2023; Costin et al., 2023), this study focuses on developing an experimental prototype to validate the proposed BIM-enabled digital twin framework for SHM.

More specifically, this paper considers the deformations of critical

Table 1
The tools used in this study.

Task	Sub-task	Tool	Description
Digital asset development	Digital structural model generation Digital model structural analysis	Robot Structural Analysis Professional 2023	A software tool that is used for structural design, structural analysis, and code compliance checking with integrated workflows to exchange data with Revit
Physical asset development	Physical structural model prototyping	6063-T52 aluminum bar	A lightweight and ductile metallic alloy used to construct the building frame prototype
		MIG welding machine	A Metal Insert Gas welding (MIG) machine that is used to connect all aluminum bars to form a building frame with rigid connections
Data Sensing, Preprocessing, Validation, and Transmission	Smart sensing and processing unit development	Raspberry Pi 4	An affordable and small-sized single board computer to control sensors for data collection and perform data processing
		Python 3	An open-source programming language that is used to develop programs for controlling the sensors to collect data
		BF350-3AA strain gauge sensor	A resistive foil that has 350 Ω nominal resistance to sense the strain by measuring the change in sensor resistance when a load is applied
		Teyleten Y3 strain gauge amplifier	A low-noise electronic device to amplify and condition the low-level signals generated by the strain gauge sensor
		TOF10120 laser distance sensor	A time-of-flight laser distance measurement device to collect distance information with a measurement range between 50 and 1800 mm
	Digital signal preprocessing and validation	Python 3	An open-source programming language that is used to develop programs for preprocessing the collected sensor data and performing data validation using the given metrics and ground truth values
	Wireless communication unit development	HTTP webserver	A software application that runs on a computer (e.g., Raspberry Pi 4) to handle Hypertext Transfer Protocol (HTTP) requests from clients (typically web browsers) and delivers data (e.g., digitals, and texts) to clients
	Wireless sensor data transmission	TL-WDR5620 router	A networking device that supports dual frequency bands and multiple antennas to enable connections between computer networks and provide wireless internet access to the devices within the network
BIM-enabled SHM Digital twin Creation and Visualization	Digital model adoption	Autodesk Revit 2024	A BIM software tool widely used in the architecture, engineering, and construction industries to enable professionals to design, visualize, simulate, and collaborate on building projects
	Domain knowledge-based color coding	Dynamo plugin for Autodesk Revit 2024	A visual programming tool that allows Python programming to request and process the sensor data from the HTTP webserver and perform real-time sensor data visualization on the BIM model

structural members of an aluminum frame prototype. The shape/geometry of the considered physical prototype is a reduced/small version (with a scaling ratio of 1:10) of a frame commonly used for modeling buildings such as industrial buildings for factories and warehouses and high-rise residential buildings. It is worth noting that this is a common practice in the field of SHM as plenty of previous studies have developed similar experimental structural prototypes for validating their IoT-enabled SHM systems (Ibanez et al., 2019; Nuzzo et al., 2021; Chiachío et al., 2022; Zerbino et al., 2023; Chang and Shirazi, 2023; Teng et al., 2023).

Fig. 3 shows a sequence of steps followed for constructing the physical structural frame, where a MIG welding machine was used (i.e., all 12 structural elements are rigidly connected by welding). The physical prototype in this study was built in the Makerspace facility located in the New Jersey Institute of Technology (NJIT) which is a manufacturing facility offering access to an array of state-of-the-art equipment for design, prototyping, testing, and research.

As shown in Fig. 3, the major process includes obtaining the raw aluminum samples, sample preparation, and aluminum welding.

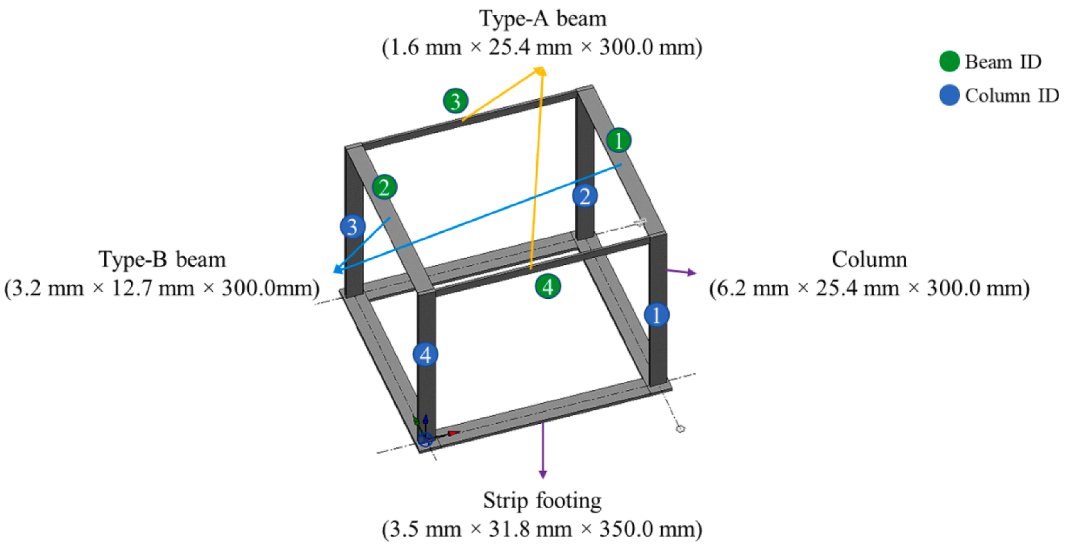


Fig. 2. Digital building frame prototype.

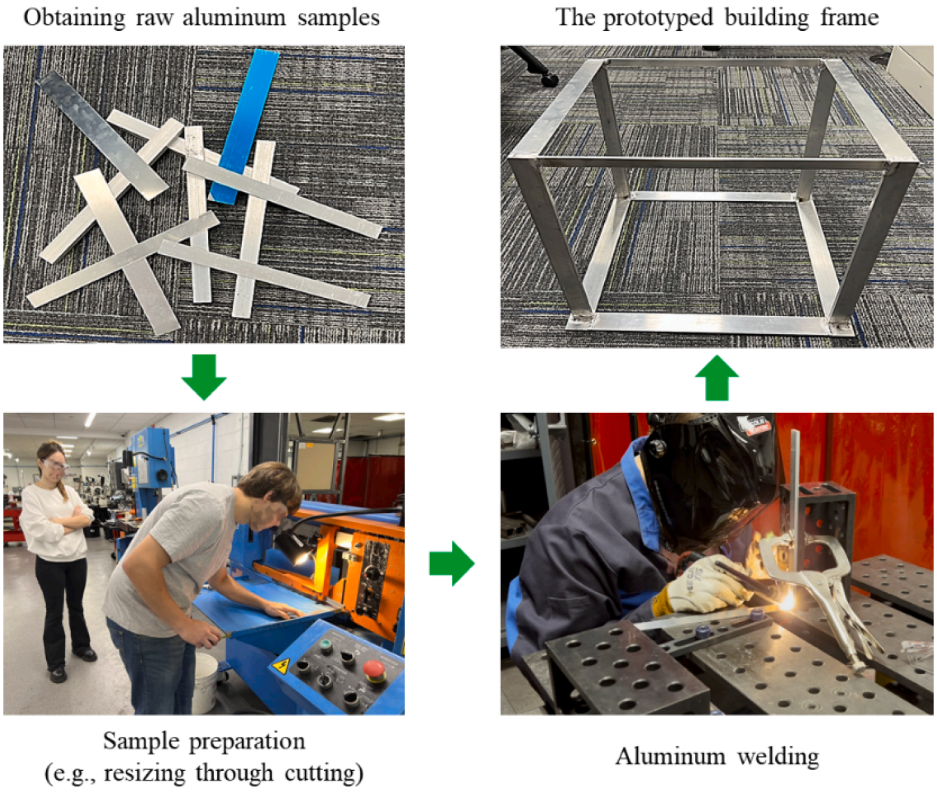


Fig. 3. Physical structural model prototyping process.

3.3.2. Smart sensing and processing unit development

Since the proposed framework focuses on monitoring structural deflection and strain, an IoT sensing unit with a single board computer (SBC) and two types of sensors was developed to enable the smart sensing of desired structural deformations. The used hardware included Raspberry Pi 4 as the SBC, BF350-3AA as a strain gauge sensor, Teyleten Y3 as a strain gauge amplifier, and TOF10120 as a laser distance sensor to collect and preprocess the strain values and distance measurements. The sensors are installed on the structure based on the optimal sensor location information obtained from the structural analysis conducted in the previous module (i.e., Module 1). The involved software tools included the use of Python 3 to develop and run programs on Raspberry Pi 4 for sensor data collection and automatic data preprocessing. More details are provided in the next subsections.

3.3.3. Digital signal preprocessing

Preprocessing the raw digital signals are critical steps to improve and validate the quality of the signals by removing noises and to ensure that the preprocessed sensor data can be used for subsequent analysis (Mousavi et al., 2017; Wang et al., 2016). In SHM, the effect of noises on strain gauge sensor signals is relatively low and can be minimized when a shielded cable and a low-noise amplifier are used (Štubian et al., 2020). A low-noise amplifier is an electrical device to amplify the weak strain gauge signals for signal strength improvement. In this study, both

shield cables and low-noise amplifier are used in strain monitoring process.

Yet, laser sensor data often carry inevitable noises due to multiple reasons such as ambient light, object reflectivity, interference from other electronics (Jia et al., 2019; Khan et al., 2020; Wang et al., 2021). Therefore, to preprocess the collected laser distance sensor data in this paper, two commonly utilized signal denoising methods were used and compared in this paper: Wavelet Transform (WT) and Fast Fourier Transform (FFT). WT dissects a signal into wavelet coefficients through convolution with a set of wavelet functions and then employs thresholding on these coefficients; hence, eliminating noise components while preserving the signal components (Silva et al., 2006). On the other hand, FFT converts a signal from the time domain to a frequency domain, and thus revealing diverse frequency spectrums. To filter out noisy spectrums, a power spectrum threshold is applied. Finally, the Inverse FFT converts the filtered frequency domain signal back to its time domain (Zhang et al., 2013).

It is of great importance to know the signal-to-noise ratio (SNR) of a signal prior to its processing because SNR is often a reliable metric to assess the quality of the signal, where a signal with high SRN is desirable for better signal quality (Fiete and Tantalo, 2001; Liu et al., 2015). This study applies the reciprocal of the coefficient of variation (which is the ratio of mean to the standard deviation of the laser distance signal) to quantify the SNR of the laser distance signal (Johnson, 2006), as shown in Eq. (1).

$$SNR = \frac{\mu}{\sigma} \tag{1}$$

Where μ and σ are the mean and standard deviation of the signal values, respectively.

Different parameters of each denoising method are investigated in order to seek an optimal denoising method to handle the noises in the laser distance signals, as summarized in Table 2.

Table 2 shows three parameters, including the decomposition level, wavelet coefficient threshold, and wavelet kernel type, which are investigated and fine-tuned for the WT method. Decomposition level refers to the number of times the distance signal is decomposed using a wavelet transform. Higher decomposition levels provide more detailed information about the signal at different scales but also result in more coefficients and a more complex representation (Jang et al., 2021). Wavelet coefficient threshold is boundary value used to suppress or eliminate wavelet coefficients that are identified with noises. Wavelet kernel type is a specific wavelet function or filter that splits the signals into components with different frequency and then studies each component with a resolution that fits its scale (Graps, 1995). This paper focuses on the use and comparison of Symlets 4 (sym4) wavelet and Daubechies 1 (db1) wavelet due to their wide applications in WT-based signal preprocessing (Song et al., 2021; Li et al., 2022).

One parameter (i.e., power spectral density threshold) of FFT is fine-tuned in this study. Power spectral density is a measure that describes how the power of a signal is distributed across different frequencies. When a signal has been transformed from time domain to frequency domain, its frequencies with power spectral density less than a threshold are identified with noises, and vice versa.

The investigated values for the parameters of two signal denoising methods are empirically determined through a trial-and-error process by beginning with a large range of values. Then, the value range for each parameter was further narrowed down to a smaller range with a finer step size when the denoised signals were observed to be close to the desired signals without noise.

Table 2
Parameter and the investigated values for distance signal denoising.

Signal denoising method	Parameter	Investigated value
Wavelet Transform (WT)	Decomposition level	{3, 4, 5, 6, 7}
	Wavelet coefficient threshold	{15, 20, 25, 30}
	Wavelet kernel type	{"sym4,"db1"}
Fast Fourier Transform (FFT)	Power spectral density threshold	{60, 80, 100, 120, 140}

3.3.4. Wireless communication unit development and sensor data transmission

A TL-WDR5620 router was used to connect the sensors to other computer networks and establish wireless communication between multiple devices. To facilitate the smooth transmission of data from the smart sensing unit to BIM software, a wireless communication unit is developed using HTTP webserver and wireless local area network (WLAN). The HTTP webserver, operating on the Raspberry Pi with internet connectivity, functions as a software application. The webserver's role involves receiving and managing periodical HTTP requests (e.g., at a requesting rate of 5 Hz) from clients, such as personal computers running BIM software. This server can deliver various resources, including web pages, files, and data, to clients within the WLAN through a request-response mechanism (Cardellini et al., 2002). In the proposed framework, one HTTP server is established for high-speed wireless communication to enable sensor data transmission.

3.4. Module 3: BIM-enabled SHM digital twin creation and visualization by integrating structural analysis knowledge

The proposed framework reuses the digital structural model developed in the digital asset development module (i.e., Module 1) as the base to create the SHM digital twin in a BIM environment. In this study, Autodesk Revit 2024 was used as the BIM-enabled visualization environment to directly adopt the structural model from the Robot Structural Analysis Professional 2023 and visualize the structural behavior and performance. It is used to create the SHM digital twin of the prototype by pulling the sensor data periodically from the webserver in a real-time manner.

To visualize deformations, a visual programming program is developed through the Dynamo extension in Revit, enabling clear representation of these changes. More specifically, a Dynamo plugin was created as a visual programming tool in Revit to request and process the sensor data from the HTTP webserver and send commands to Revit for performing the desired real-time sensor data visualization on the BIM model.

The structural deformations under various loadings are monitored by innovatively integrating the sensor data and the existing domain knowledge of structural mechanics and analysis to effectively interpolate strain deformation data obtained from a single sensor. For structural health evaluation and visualization purposes, the interpolated deformation values are color coded based on pre-defined limit states. Two load configurations are utilized to induce deformation in the beam: a concentrated load $\boldsymbol{P}$ applied at the midspan, and a distributed load $\boldsymbol{w}$ applied along the entire beam. The beams for these two different load configurations have both ends fixed supports due to their rigid connection with columns in the prototyped structural frame. Additionally, a lateral load is applied to induce deformation in the column. More details are provided in the next subsections.

3.4.1. Load configuration #1: Concentrated load at beam midspan

Fig. 4 presents: (1) the schematics of beam #1 (see Fig. 2) with a concentrated load applied at the midspan, and (2) the corresponding linear bending moment diagram which is a fundamental relationship in the domain of Engineering Mechanics (Malesza, 2013). The strain gauge sensor is attached at the midspan on the side experiencing positive moments due to the tensile bending. It is positioned to capture the peak tensile strain specifically on the bottom surface (also known as the lowermost fiber in structural terms) of the section.

As shown in Fig. 4, the targeted beam has a length of L in the x direction (which is 300.0 mm for the developed prototype). In any cross-section normal to the x direction, the maximum tension strain occurs on:

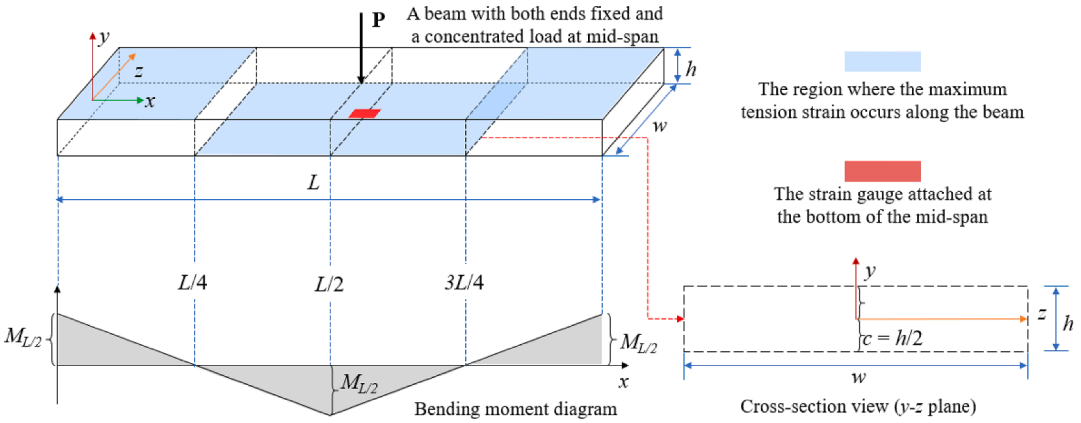


Fig. 4. Schematics and bending moment diagram of a beam under concentrated load.

(1) the top surface if x is less than $L/4$ and greater than $3L/4$, and (2) the bottom surface where x is between $L/4$ and $3L/4$. Meanwhile, the bending moment diagram resulting from the concentrated load follows a linear pattern. This means a consistent linear relationship is maintained across the beam's length for all cross-sections.

Due to fixed support at both ends, the boundary conditions for the beam under load configuration #1 should meet the requirements that the transverse displacements (δ) and rotations (θ) are zeros at both ends ($x = 0$ and $x = L$), which are given in Eq. (2) and Eq. (3), respectively. This is because the fixed ends prevent any translation or rotation at those end points. The same boundary conditions apply for the beam in load configuration #2 (as shown in the next subsection) as the beam also has both ends fixed supports.

$$\delta = \begin{cases} 0, x = 0 \\ 0, x = L \end{cases} \tag{2}$$

$$\theta = \begin{cases} 0, x = 0 \\ 0, x = L \end{cases} \tag{3}$$

For loading configuration #1, the maximum bending moment of beam #1 occurs at the midspan and each end of the beam with the same magnitude which is represented as $M_{L/2}$. Therefore, in conjunction with the bending moment diagram in Fig. 4, the bending moment at any location in the x direction can be mathematically represented in terms of $M_{L/2}$, as shown in Eq. (4).

$$M_x = \begin{cases} M_{L/2}\left(1 - \dfrac{4x}{L}\right), 0 \le x \le L/2 \\ M_{L/2}\left(\dfrac{4x}{L} - 3\right), L/2 < x \le L \end{cases} \tag{4}$$

According to the strain distribution theory for pure bending, the maximum tension strain ($\varepsilon_{L/2}^{max}$) at the midspan can be expressed using Eq. (5).

$$\varepsilon_{L/2}^{max} = \frac{M_{L/2} \bullet c}{IE} \tag{5}$$

Where x is the location of a cross-section along the x direction; I represents the second moment of area (moment of inertia); E denotes Young's modulus; c is the extreme distance from the centroid of the cross-section to the top or bottom surface. Since c, I, and E are constants, the maximum tension strain at midspan is linearly proportional to the magnitude of bending moment.

By combining Eq. (4) and (5), the maximum tension strain for any cross-section (ε_x^{max}) along the beam can be interpolated using Eq. (6) when the maximum tension strain at the midspan cross-section is known. Specifically, $\varepsilon_{L/2}^{max}$ is directly measured in this paper by a strain gauge sensor that is attached at the midspan on the bottom surface of the beam.

$$\varepsilon_x^{max} = \begin{cases} \varepsilon_{L/2}^{max}\left(1 - \dfrac{4x}{L}\right), 0 \le x \le L/2 \\ \varepsilon_{L/2}^{max}\left(\dfrac{4x}{L} - 3\right), L/2 < x \le L \end{cases} \tag{6}$$

3.4.2. Load configuration #2: Uniform distributed load on the beam

Fig. 5 presents: (1) the schematics of beam #1 (see Fig. 2) with a uniform loading applied along its length, and (2) the corresponding quadratic bending moment diagram that is widely acknowledged in the domain of Engineering Mechanics. As in the case of concentrated loading (i.e., see previous subsection), the strain readings are taken from the strain gauge sensor attached at the midspan on the side experiencing positive moments due to the bending. It is positioned to capture the peak tensile strain specifically on the bottom surface of the section.

As shown in Fig. 5, in any cross-section along the beam length, the maximum tension strain in each cross section occurs on: (1) the top surface if x is less than $(\sqrt{3}-1)L/2\sqrt{3}$, and greater than $(\sqrt{3}+1)L/2\sqrt{3}$, and (2) the bottom surface otherwise. Meanwhile, the magnitude (M_x) of the bending moment in a cross section of the beam along its x axis has a quadratic relationship with the location of the x, as represented by Eq. (7). In addition, the bending moment of beam at the midspan and each end of the beam is represented as $M_{L/2}$., M_1, and M_2, respectively, where $M_1 = M_2$. For a given uniform distributed load ($\boldsymbol{w}$) and the length of the beam (L), The values for $M_{L/2}$. , M_1, and M_2, can be derived using Eq. (8). Therefore, the bending moment at any cross-section x along the beam length can be formulated as shown in Eq. (9).

$$M_x = ax^2 + bx + c \tag{7}$$

$$\begin{cases} M_1 = M_2 = \dfrac{w \bullet L^2}{12} \\ M_{L/2} = -\dfrac{w \bullet L^2}{24} \end{cases} \tag{8}$$

By solving Eq. (7) for $M_{L/2}$, M_1, and M_2 values given by the relations in Eq. (8) corresponding to $x = 0, L/2, L$, the bending moment at any location in the x direction can be calculated using the bending moment at the midspan ($M_{L/2}$), as shown in Eq. (9).

$$M_x = \left(\frac{12}{L^2}x^2 - \frac{12}{L}x + 2\right) \bullet M_{L/2} \tag{9}$$

By combing Eq. (5) and Eq. (9), the maximum tension strain for any cross section (ε_x^{max}) along the beam can be interpolated using the Eq. (10). In the similar manner, $\varepsilon_{L/2}^{max}$ can be directly measured by a strain

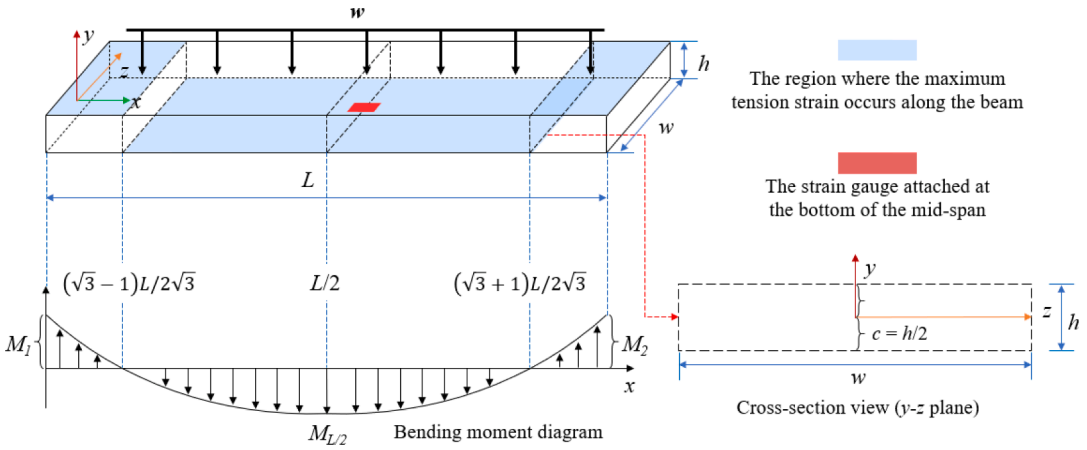


Fig. 5. Schematics and bending moment diagram of a beam under distributed load.

gauge sensor that is installed at the midspan on the bottom surface of the beam.

$$\varepsilon_x^{max} = \left(\frac{12}{L^2}x^2 - \frac{12}{L}x + 2\right) \bullet \varepsilon_{L/2}^{max} \tag{10}$$

4. Results and analysis

This section presents and analyzes the obtained results.

4.1. Digital structural model with structural analysis results

Figs. 6 and 7 visualize the digital representations of the structural frame under two different loading configurations along with the conducted structural analysis to identify the critical locations where large deformations occur.

As shown in Fig. 6(a), the vertical concentrated loads are applied incrementally at the midspans of the beams #1 and #2. These two beams are more prone to bending deformations due to lower second moment of area in the direction where load is applied, as opposed to beams #3 and #4. To stimulate the deflection of the columns, a dynamic lateral concentrated load is applied at the midspan of beam #3. As a result, bending moment diagrams for beams #1 and #2 and the four columns are generated automatically through structural analysis, which is shown in Fig. 6(b).

Due to the fixed-end supports, the orientation of the bending moment is not consistent along the beam. Using beam #1 as an example, on the beam surface where the concentrated load is perpendicular, the load induces negative stress/strain in the beam cross-sections that are in parallel to the load and close to each end. This structural behavior indicates that the bottom half of the beam cross-sections is subject to compression while the top half experiences tension. For other cross-sections that are close to the midspan, the bottom half of those cross-sections are under tension while the top half is subject to compression. The bending moments that induce tension and compression are represented by light red and blue, respectively, as shown in Fig. 6(b).

Disregarding the orientation of the bending moment, for beam #1, the maximum bending moment occurs at three locations in beams: at the midspan and the two ends. As the maximum bending stress/strain on a cross-section of beam carries the largest bending moment, therefore, the maximum bending strain appears in the cross-sections at the midspan and the two ends for this case. Accordingly, strain gauge sensors should be placed at the midspan or the ends of the beam so that the sensor can

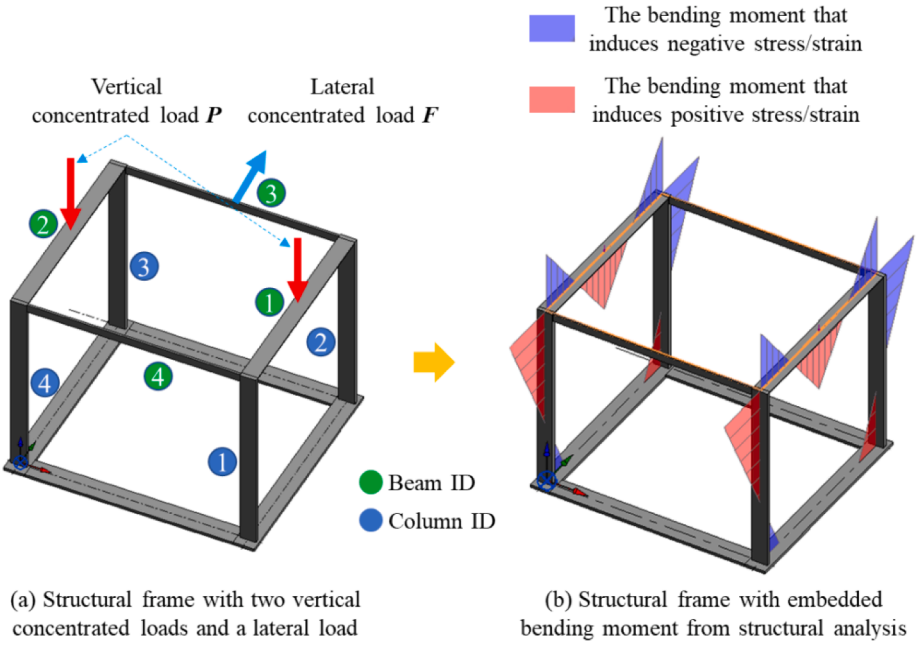


Fig. 6. Structural model and analysis with concentrated loads and a lateral load.

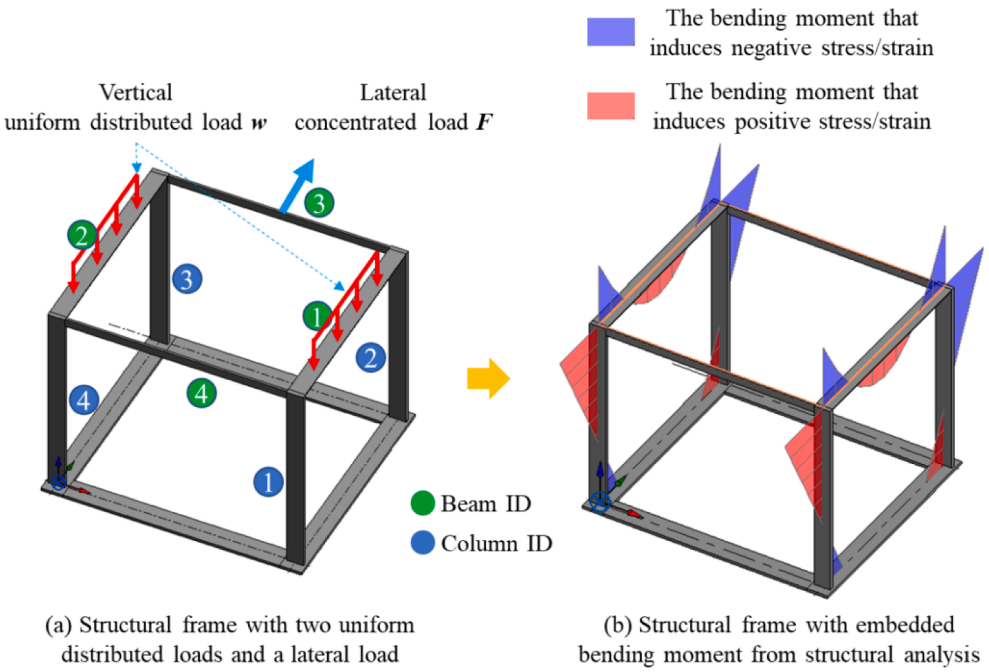


Fig. 7. Structural model and analysis with distributed loads and a lateral load.

measure the maximum strain in the beam. Hence, this study places a strain gauge sensor at the bottom surface of beam #1 to measure the maximum tension strain for beam strain distribution monitoring.

Similarly, Fig. 7(a) displays: (1) the digital model with uniform distributed loads applied on beams #1 and #2, and (2) a lateral concentrated load applied on beam #3.

The bending moment diagram of the two beams and four columns are shown in Fig. 7(b). For beams #1 or #2, the maximum bending moment occurs at the ends of the beams. However, in practice, it is infeasible to place strain gauge at the ends due to issue of mounting and fixturing caused by interference with the beam's end supports (Kim et al., 2013). Therefore, the strain gauge is placed at the bottom of the midspan as the second largest bending moment occurs there. The tension strain measurement at the location where the second largest bending moment occurs is used to interpolate the tension strain at other locations for strain visualization purposes.

Both Fig. 6(b) and Fig. 7(b) indicate that the maximum bending moment in any of the four columns occurs at the top of the column from the footings. As a result, the maximum deflection also happens at the top location of the columns. Thus, the laser distance sensor is placed at the same elevation of the column top.

4.2. *Physical structural model with installed smart sensing unit*

Given the constructed physical structural frame and the optimal sensor installation location determined through structural modeling and analysis (see previous subsection), a smart sensing unit was constructed and installed on the physical frame, which is shown in Fig. 8. Specifically, this smart sensing unit relies on sensor technologies to allow accurate and automated collection of data that is further processed by an SBC (i.e., Raspberry Pi 4B).

Fig. 8 presents the overview of the physical structural model with installed smart sensing unit. Specifically, a BF350-3AA strain gauge sensor is attached on the bottom surface of beam #1 at its midspan (see

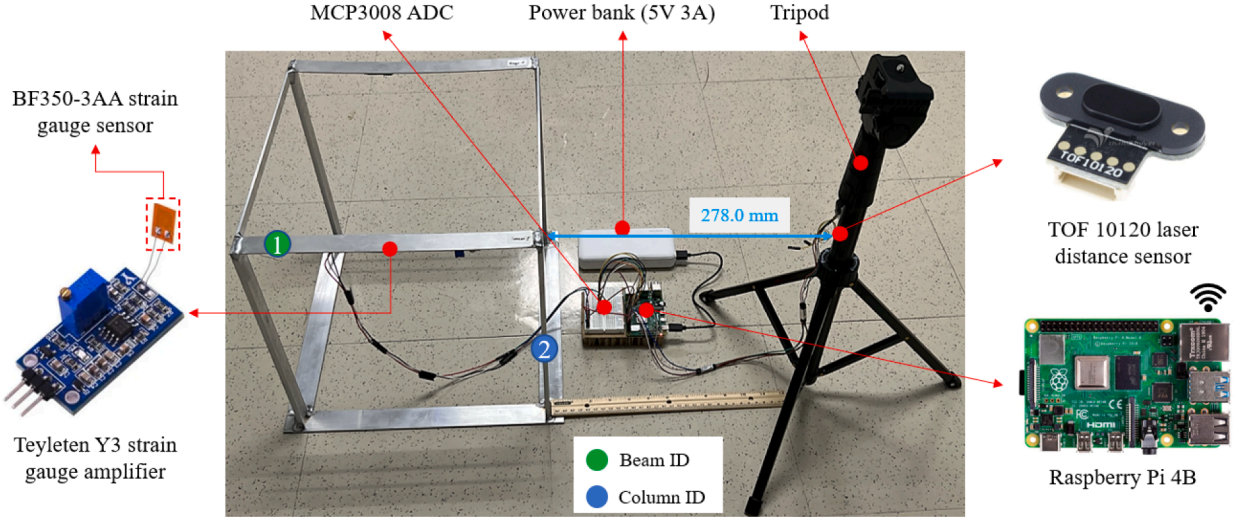


Fig. 8. The sensing unit installed on the prototyped physical structural model.

Fig. 8). This gauge sensor also connects to the Teyleten Y3 strain gauge amplifier which communicates with the SBC through an MCP3008 analog-to-digital converter. This converter enables the conversion from digital values to analog values (i.e., continuous range of digits) so that the strain values can be obtained. The SBC is powered by a power bank with a 5 V-3A output, and the other two sensors are indirectly powered by the power bank through this SBC. Meanwhile, a TOF 10120 laser distance sensor is placed at an elevation of 300.0 mm with a fixed perpendicular distance of 278.0 mm (see Fig. 8). A fixed distance of 278.0 mm is chosen as it falls between the range of 100 mm and 1,800 mm within which the selected laser distance sensor has the highest sensing precision (Bonaldo et al., 2021).

4.3. Optimal method for laser distance signal preprocessing

Fig. 9 shows a sample of raw distance signal collected by the TOF 10120 laser distance sensor, in support of developing an optimal signal denoising method for laser distance signal preprocessing.

As shown in Fig. 9, the raw distance signal is comprised of distance measurements collected during three sub-periods: 0 ~ 90 s, 91 ~ 315 s, and 316 ~ 420 s. In particular, the signals from 0 ~ 90 s and 316 ~ 420 s contain static distance measurements, which means that the distance information is collected when both the physical structure and laser distance sensor remain stationary. However, the signals from these two periods fluctuate up and down and show great variance of the distance measurements, ranging from 270 mm up to 290 mm. Compared to the actual fixed distance of 278 mm, the absolute distance measurement error is about 8 ~ 10 mm, causing a low distance measurement precision if a millimeter-level deflection is desired. Regarding the signal during 91 ~ 315 s, it shows an obvious bell-shape pattern when lateral load is applied and released periodically, demonstrating the effectiveness of the sensor for distance measurement. Meanwhile, the signal segments for the static distance measurements have SNR of 67.19 and 71.53, which are much greater than that of the dynamic distance signal (i.e., 29.58). This further indicates that the quality of the static distance signal segments is higher than that of the dynamic distance signal segment. The entire signal has low smoothness with an overall low SNR of 29.56 due to the noises caused by ambient light, object reflectivity, and interference from other electronics. This further shows and validates that an optimal signal processing method is required to clean the laser distance signal by removing the noises so that the signal becomes smooth to better represent the actual distance measurements.

Fig. 10 ~ 12 show the raw distance signal as well as the preprocessed distance signals using both FFT and WT algorithms with different parameters.

As shown in Fig. 10, one significant observation is that the entire preprocessed signal tends to be smoother when the power spectral density threshold (PSDT) increases from 60 to 100 V^2/Hz but has minor changes in smoothness with a higher PSDT than 100 V^2/Hz. In addition, regardless of the PSDT, FFT performs well to smooth the signal (i.e., dynamic distance measurement) with lower SNR. For the high-SNR distance signal segments that represent static distance measurements, FFT could remove some noises as the preprocessed signal segments are smoother compared to the original signal segments. Therefore, the FFT denoising method performs the best for distance signal denoising with a PSDT of 100 V^2/Hz. However, the preprocessed signal segments for the high-SNR signal segments are still subject to instability because the distance measurements show a wavy rather than a flat pattern in the preprocessed signals. This observation further reveals that both the PSDT and SNR would affect the performance of FFT on laser distance signal denoising. FFT can particularly well denoise the signals with low SNR yet suffer from lower performance on handing the noises in the high SNR signals.

Figs. 11 and 12 display the raw distance signal and the preprocessed distance signals by WT using 'db1′ kernel and 'sym4′ kernel with other different parameters, respectively. A series of trial experiments show that wavelet coefficient threshold (WCT) of 25 works the best for both the 'db1′ kernel and 'sym4′ kernel. Therefore, the WCT is fixed as 25 when investigating the impact of another parameter (i.e., signal decomposition level) on the performance of the WT with any of these two kernels.

Fig. 11 shows two major observations. First, regardless of the signal decomposition level (DL), a WT with 'db1′ kernel generates a preprocessed signal with ladder-like patterns. The insight is that 'db1′ (i.e., Daubechies 1) is a simple wavelet that resembles a step function, which has a compact support and is particularly effective at representing and detecting abrupt changes or step-like features in signals (Daubechies and Sweldens, 1998). Due to this unique characteristic, for all preprocessed signals, WT with 'db1′ kernel well differentiates the signals from static distance measurements to dynamic distance measurements, especially when the DL is greater than 3. Second, a higher DL enables the preprocessed data by 'db1′ kernel to be smoother with less fluctuations. For example, when the DL is at 3, the preprocessed signals for both static and dynamic distance measurements still contain plenty of step-ups and step-downs. Once the DL reaches 6, the preprocessed signal tends to be stable, which indicates that continuously increasing the signal DL may not be able to improve the performance of WT with 'db1′ kernel. Therefore, the optimal parameter for WT with 'db1′ kernel includes a WCT of 25 and a DL of 6.

Similarly, Fig. 12 shows the raw distance signal and the preprocessed distance signals by WT with 'sym4′ kernel at different signal decomposition levels.

In Fig. 12, overall, the WT with 'sym4′ kernel well handles both high-SNR and low-SNR signals, especially at a higher signal DL. For instance, a 'sym4′ kernel with higher DL results in flatter and smoother signals for the high-SNR signals without compromising the denoising performance for low-SNR signals as the preprocessed low-SNR signals have minor changes. When DL equals to 6, the WT with 'sym4′ kernel reaches the

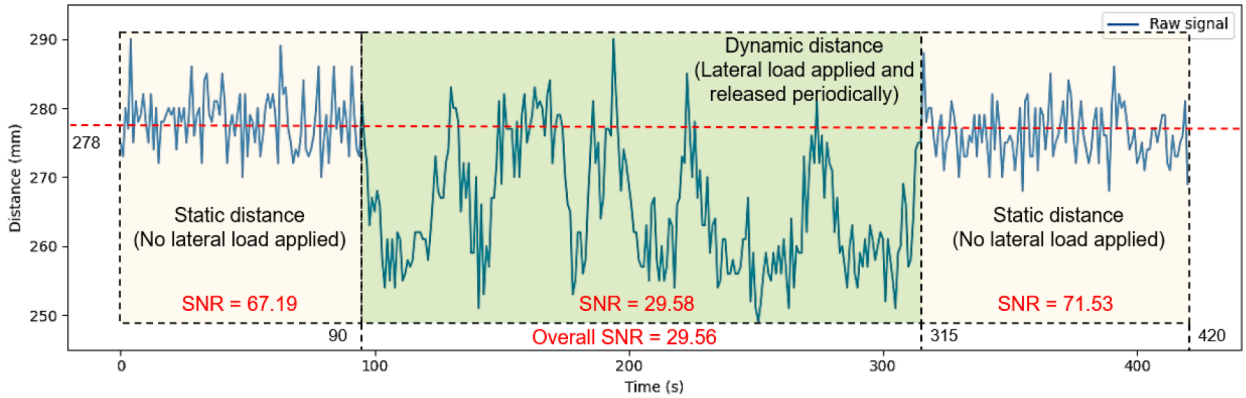


Fig. 9. Sample of raw distance signal collected.

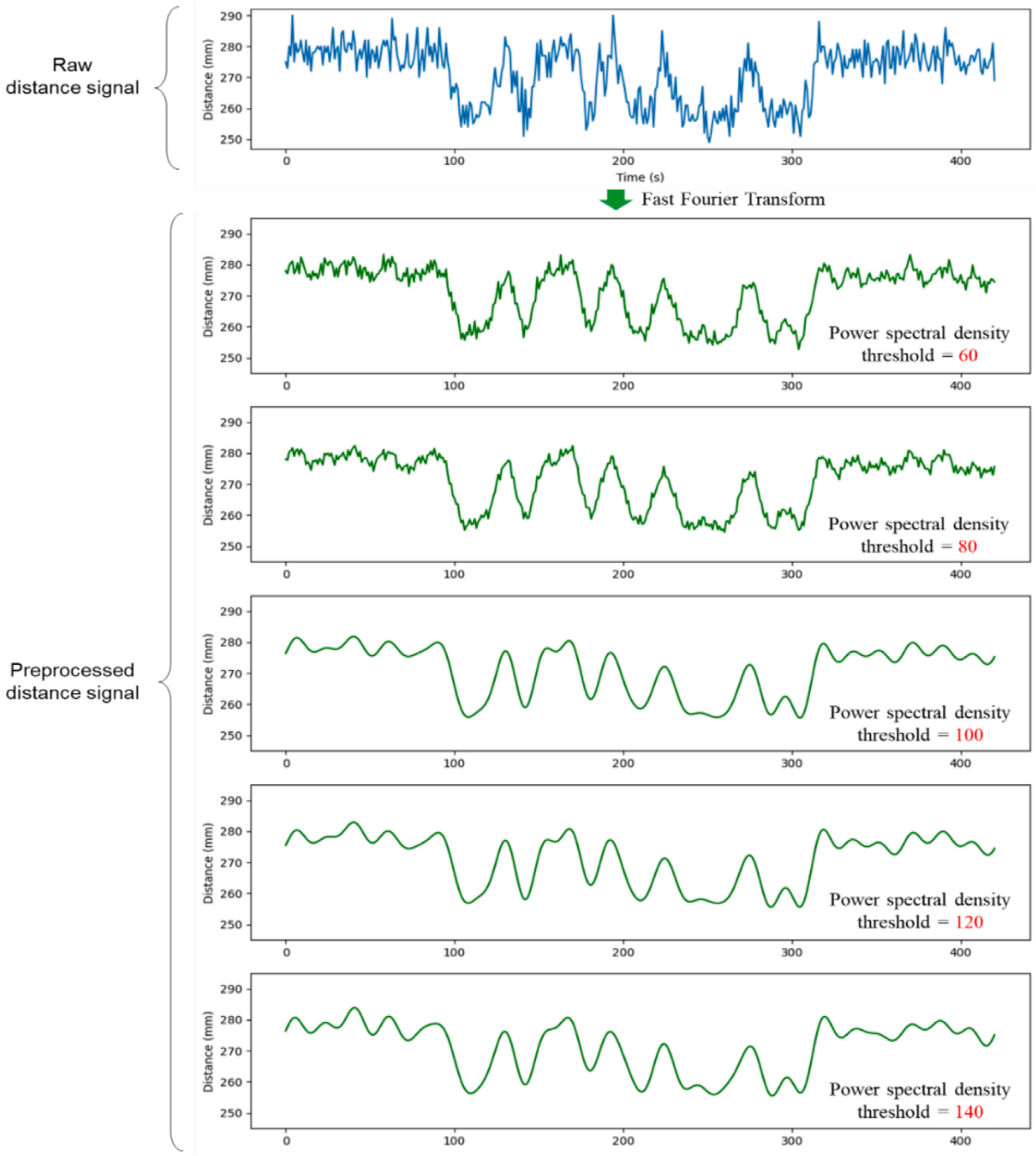


Fig. 10. Fast Fourier Transform for signal preprocessing.

best performance as continuously increasing the DL cannot further change the preprocessed distance signals. Thus, the optimal WT denoising method with 'sym4' kernel has a WCT of 25 and a signal DL at 6 as well.

Fig. 13 summarizes and compares the preprocessed signals using the signal denoising methods with different optimal parameters identified in this study for laser distance measurement.

As displayed in Fig. 13, the preprocessed signals by WT with 'sym4' kernel (WCT = 25; DL = 6) can best represent the actual static distance measurements and match the bell-shape patterns of the dynamic distance measurements when a lateral load is applied and released periodically. Therefore, it was adopted for preprocessing the laser distance signals collected by the TOF 10120 sensor by scripting and running a Python program on the Raspberry Pi 4B SBC before publishing the data onto the webserver for subsequent data analysis and visualization.

Three patterns were found by comparing the optimal WT and optimal FFT denoising methods shown in Fig. 13. First, both methods could capture the overall trends of the signals, including the flat segments denoting the static distance measurements and wavy segments representing the dynamic distance measurements. Second, FFT was more sensitive to local variations in the flat segments, resulting in residual noises in the signals for static distance measurements. Third, WT performed well to smoothen the flat segments in order to denoise the signals that can better represent the static distance measurements. However, this also exposes a potential issue that WT may suffer from the incapability of capturing the local variations that are not caused by noises, ultimately resulting in a loss of detailed information in the signals. These three patterns further indicate that (1) WT is better for

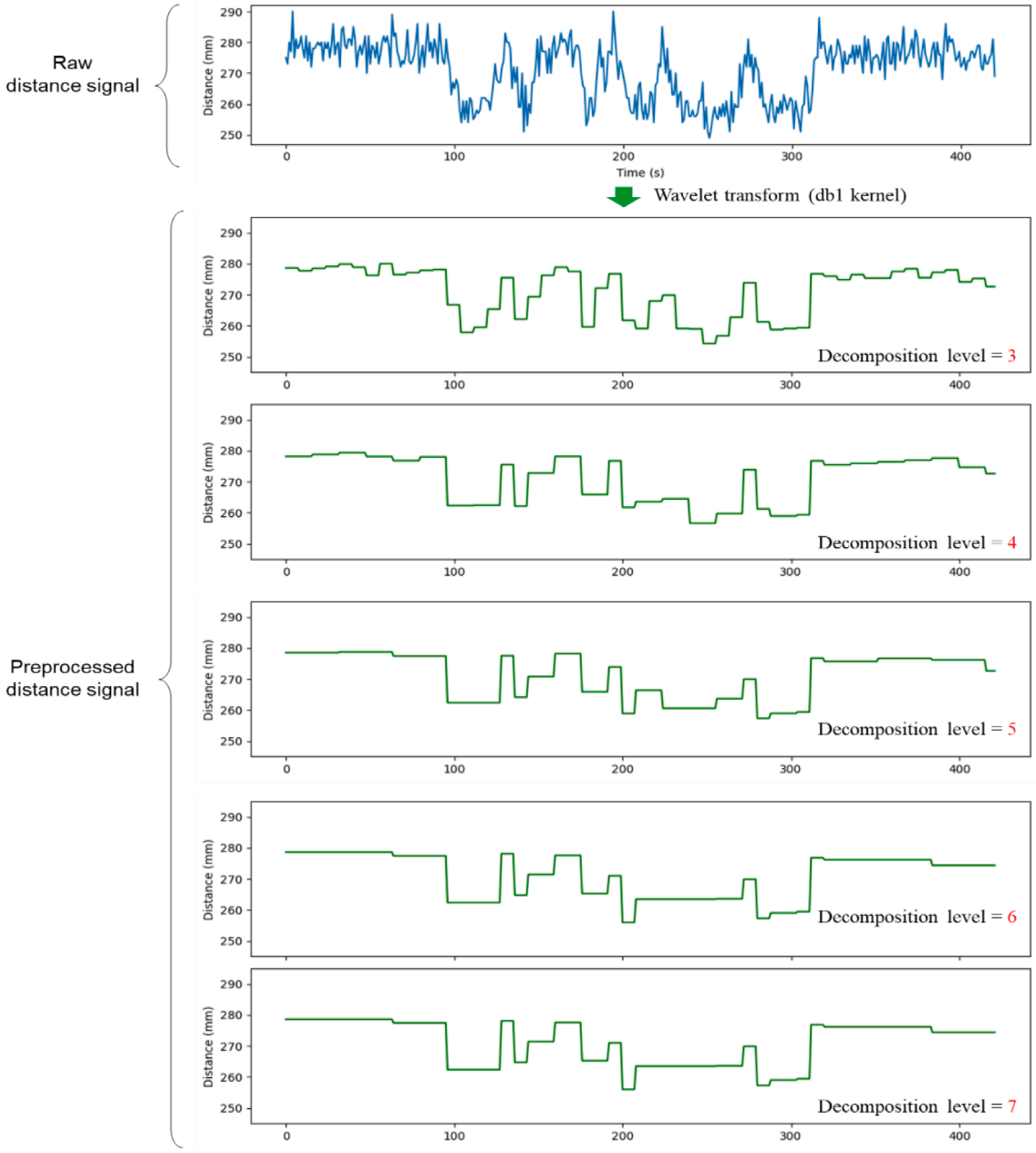


Fig. 11. Wavelet Transform ('db1' kernel; wavelet coefficient threshold = 25) for signal preprocessing.

denoising the signals to obtain a smoother signal curve without preserving much detailed information, and (2) FFT can well denoise the signals to capture the overall trend of the signals but reserve more detailed information. Therefore, FFT is more suitable for laser distance measurements that require higher precision as it can better differentiate the local variations, while WT is better for denoising laser distance signals for capturing the overall variations in the signals to get a smoother laser distance measurement.

4.4. *Dynamo graphical programming for structural behavior visualization*

Given a measured strain value, the proposed domain knowledge-guided structural behavior visualization method (detailed in the Methodology section) is able to interpolate a strain value for any cross-section that is in parallel with applied concentrated load, and then color-coding scheme can be further applied to visualize both the measured and interpolated strain. However, in Revit, it is impractical to split a beam object into unlimited cross-sections due to the limitation of the software's functionality and efficiency. Instead, it is possible to divide the beam into finite parts/cells through spitting elements. Spitting elements is one available functionality in Revit to allow a BIM modeler to cut a model element into individual parts, ultimately resulting in a set of smaller cells (Autodesk, 2024). Therefore, the digital model of beam #2 is split into multiple cells for visualizing the tension strain distribution under concentrated and uniform distributed load.

Fig. 14 shows the schematics followed by this paper to visualize the beam tension strain under a concentrated load in Revit through Dynamo graphical programming.

As shown in Fig. 14, in the x direction, 9 cells are considered (numbered from 1 to 9) and each cell has an automatically assigned element ID by Revit (e.g., cell 1 has an ID of 452861). The strain value in the cross-section where the centroid of each cell represents the average tension strain for the entire cell. Particularly, cells 2 to 8 have the same length in the x direction, while the length of cells 1 and 9 is only a half of

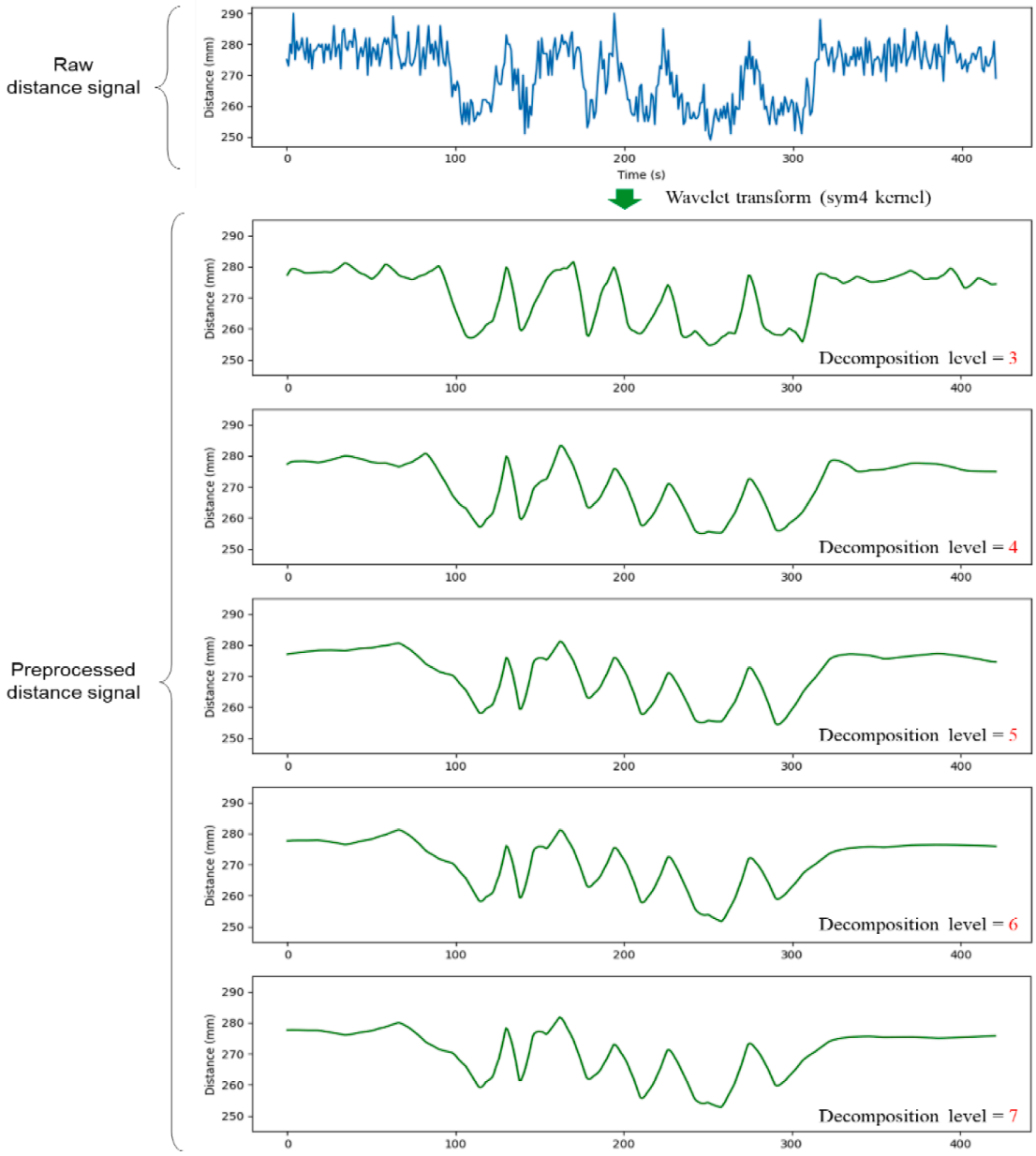


Fig. 12. Wavelet Transform ('sym4' kernel; wavelet coefficient threshold = 25) for signal preprocessing.

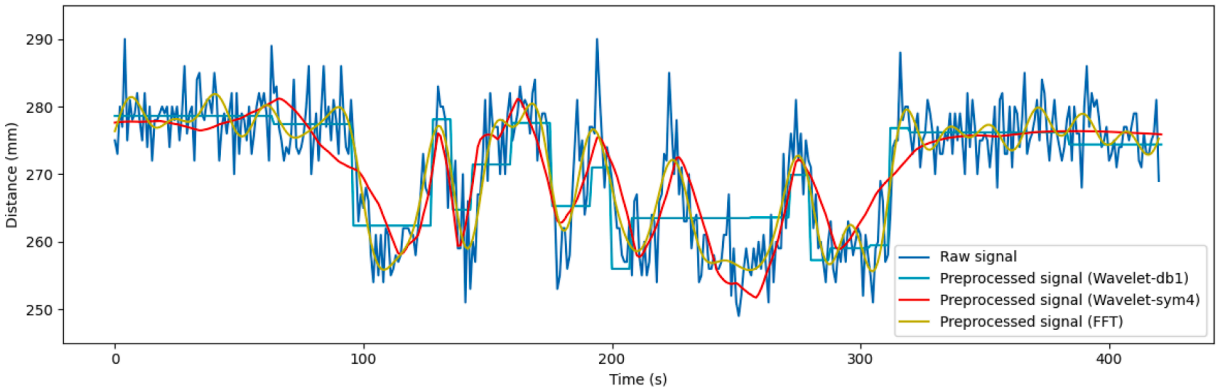


Fig. 13. Raw distance signal and the preprocessed signals by different optimal denoising methods.

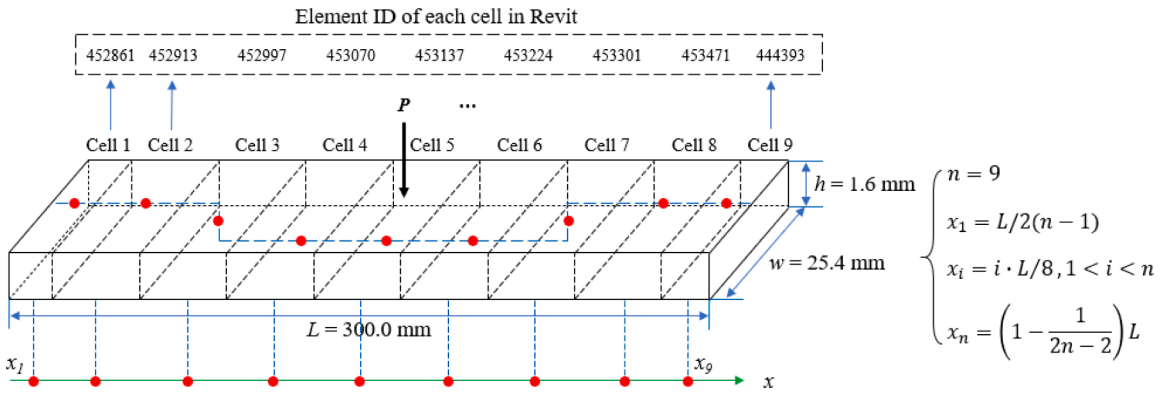


Fig. 14. Schematics of maximum beam tension strain visualization in BIM (concentrated load).

the length of any other 7 cells. Therefore, the coordinate of the centroid in the *x* direction can be derived using the formula shown in Fig. 14 given the total length of the beam. Ultimately, the sensed mid-span strain is used to interpolate the tension strain of other cells.

Similarly, Fig. 15 shows the schematics followed by this paper to visualize the maximum beam tension strain under a uniform distributed in Revit through Dynamo graphical programming. In the *x* direction, the beam is split into 7 cells that are numbered from 1 to 7 and each cell has an automatically assigned element ID by Revit. Particularly, cells 2 to 6 have the same length in the *x* direction, while the length of cells 1 and 7 is equivalent to the *x* coordinate where the cross-section separates cells 1 and 2. Therefore, the coordinate of each cell's centroid in the *x* direction can be derived given the total length of the beam. Ultimately, sensed mid-span strain is used to interpolate the tension strain of other cells.

Fig. 16 shows a sample of the developed Dynamo graphical programming interface created to request strain measurements and pre-processed laser distance from the webserver, process the requested data, and perform color coding by following the schematics in Fig. 14 to visualize the maximum beam tension strain distribution.

As shown in Fig. 16, the Dynamo interface includes five main modules to reach the desired goal. These modules are sensor data web request, sensor data processing, object ID retrieval for each cell of the beam and the targeted column, maximum column deflection visualization, and maximum beam tension strain distribution visualization. The developed Dynamical graphical program is reliably running at an optimal sample rate of 5 Hz to request and receive data from the web server and then performs real-time visualization of beam strain distribution and maximum column deflection.

4.5. Real-time beam tension strain distribution visualization

This section provides the real-time visualization for the two different load combinations considered in this paper.

4.5.1. Concentrated load condition

Fig. 17 shows three snapshots of the dynamically updated digital BIM model of the structural frame in Revit. More specifically, it shows the tension strain distribution under a concentrated load applied to the physical real-world prototype model at beam midspan when sensor data streams into this BIM software in a real-time manner.

As shown in Fig. 17, at the time stamp of t_0, no tension strain is measured as there is no concentrated load applied at the midspan; hence, the entire beam is colored uniformly green in the digital twin. As the applied load increases, at time t_2, the measured tension strain at midspan is close to the limit of the strain gauge sensor, which is a strain of 0.02 (or 2 %). As a result, the beam segment at the midspan is colored in red, while the other beam segments are also colored simultaneously depending on the measured strain value. At the same time, the two segments close to both ends of the beam are colored in light red, which means that the tension strain values in those two segments are approaching the strain limit as well. Therefore, the results of this real-time visualization well align the existing domain knowledge for strain distribution of a beam with fixed ends under concentrated load applied at midspan.

4.5.2. Uniform distributed load condition

Fig. 18 displays three snapshots of the digital structural model that is updated by the collected sensor data in real-time when the beam in the physical real-world prototype model is subject to uniform distributed loads.

Similarly, Fig. 18 also shows that increasing the density of the distributed load in real-world (i.e., in the physical prototype model) results in higher tension strain measured in the beam. As a consequence, the beam segments with higher tension strain are colored in colors close to red, and those with lower tension train are colored in colors close to green (see Fig. 18).

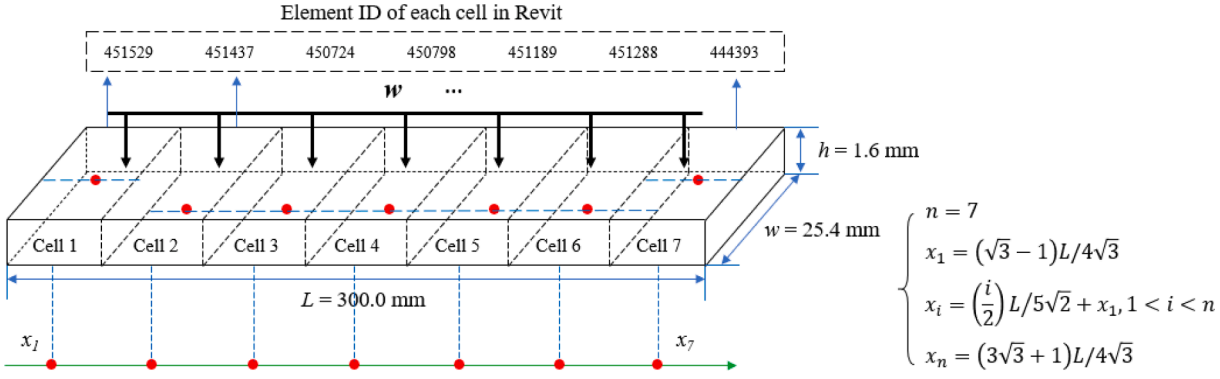


Fig. 15. Schematics of maximum beam tension strain visualization in BIM (uniform distributed load).

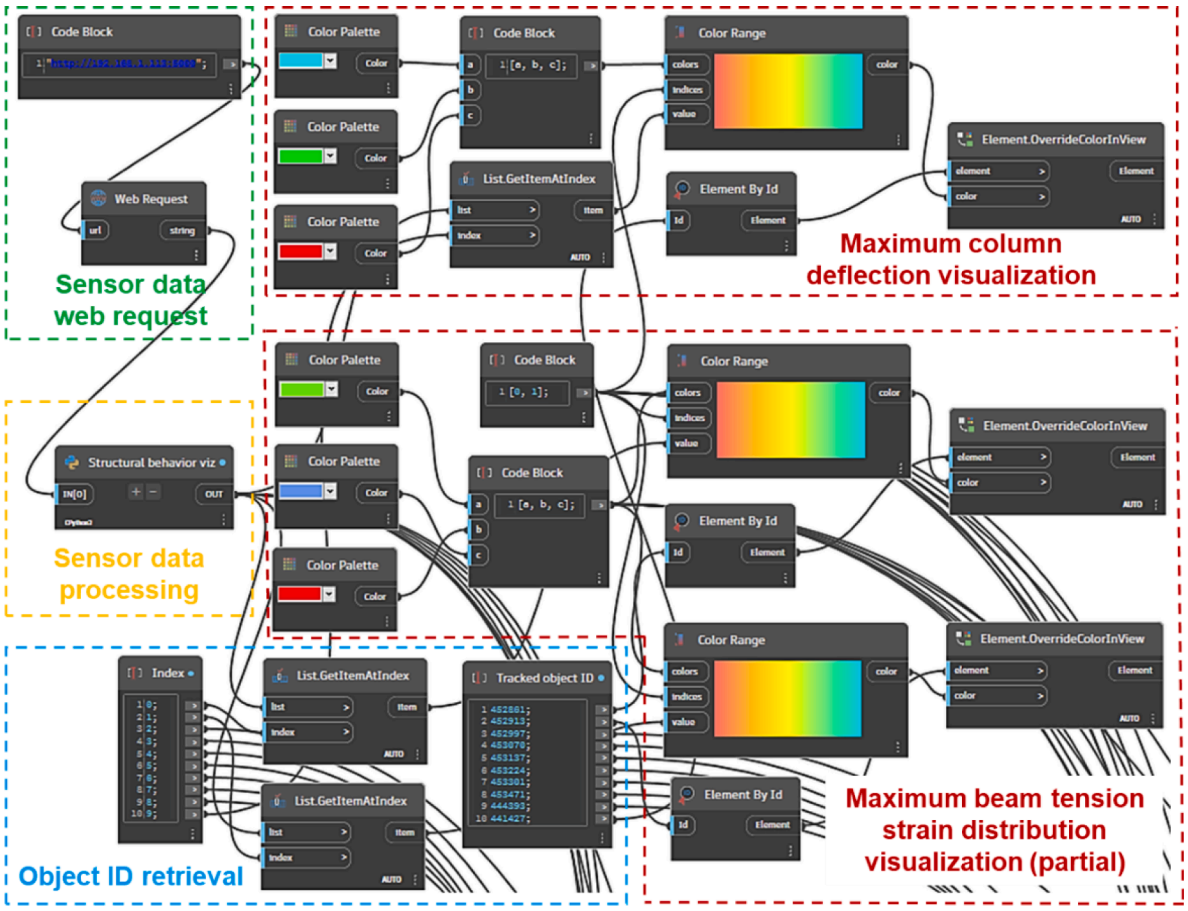


Fig. 16. Dynamo interface for maximum beam tension distribution visualization.

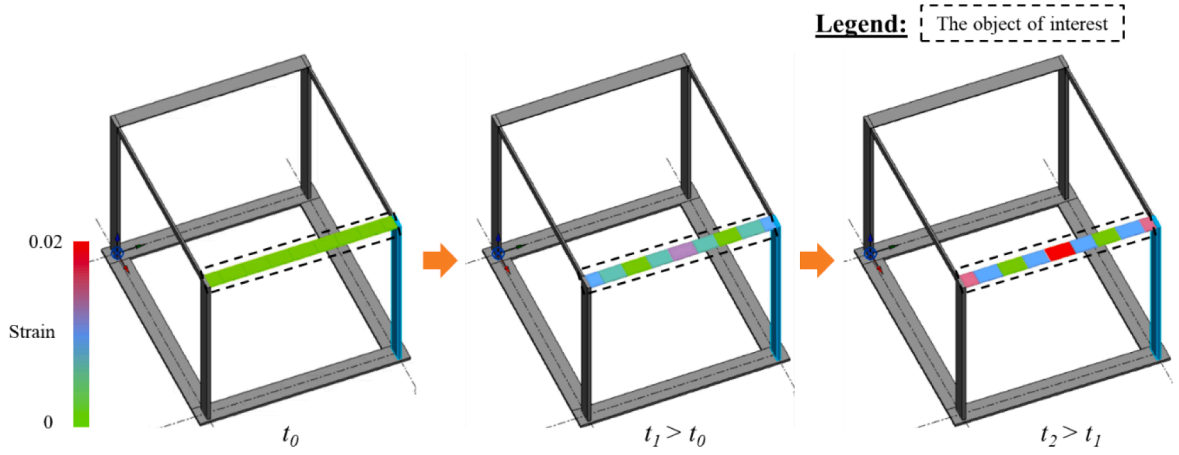


Fig. 17. Snapshots of maximum beam tension strain visualization in Revit (Concentrated load).

4.6. Real-time column deflection visualization

Fig. 19 shows three snapshots of the dynamically updating digital model of the structural frame when a lateral concentrated load is applied on the physical real-world prototype structure to stimulate column deflection.

As shown in Fig. 19, the developed smart sensing unit accurately senses the deflection caused by the lateral load in the physical world. For example, at time t_2, a lateral load results in a maximum deflection of 30 mm at the tip of the column of interest. In the digital model, the same column is colored in dark red accordingly to show the actual deflection (see Fig. 19).

5. Discussion, contributions, and limitations

5.1. Importance of high updating frequency for digital twin-based SHM systems

First of all, this study developed a digital twin-based real-time SHM system with a much higher system updating frequency than the existing similar systems. In fact, updating frequency of a real-time SHM system is a standard metric to evaluate the system's performance regarding its

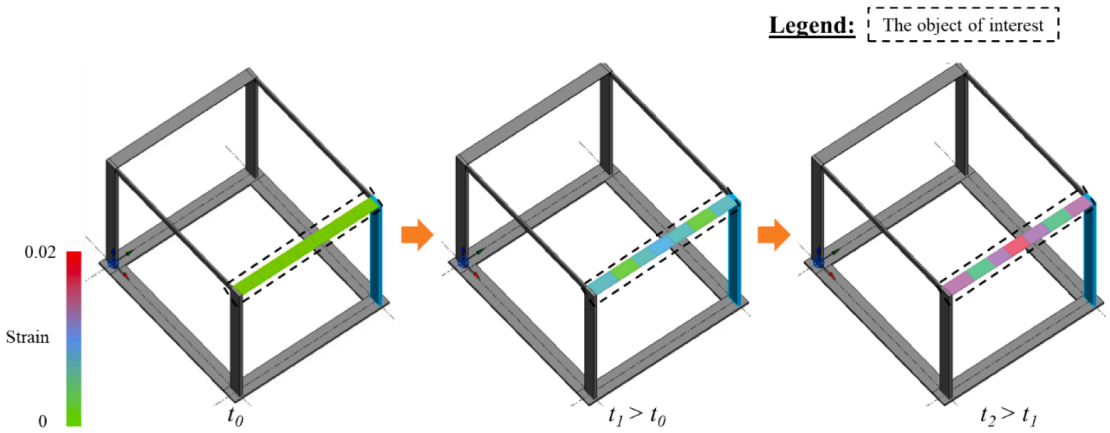


Fig. 18. Snapshots of maximum beam tension strain visualization in Revit (Distributed load).

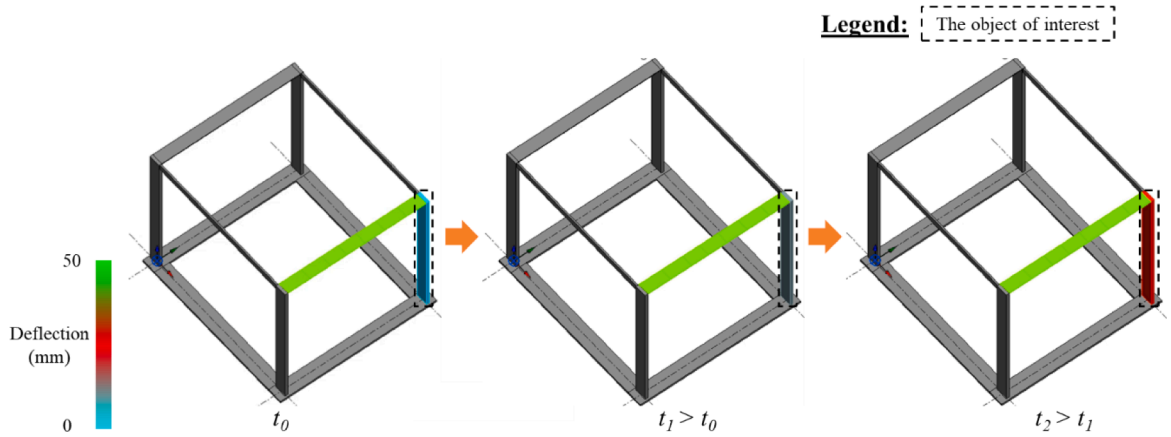


Fig. 19. Snapshots of maximum column deflection visualization in Revit (Distributed load).

capability of real-time responses as it is a packed assessment on the applied technologies (e.g., sensors, computational performance, and hardware), types and volume of data, and inference methods (Bado et al., 2022). It is worth noting that a digital twin-based SHM system is equipped with the capability of real-time data acquisition and processing, while a real-time SHM system may not be in the context of digital twins when the digital/virtual replicas of the physical objects (e.g., a bridge) are not involved. Table 3 summaries the most recent studies directed to digital twin-based and/or real-time SHM system, which are used as references to assess the proposed framework in this study based on updating frequency.

As shown in Table 3, a large proportion of the developed systems [e.g., Fawad et al (2023), and Teng et al. (2023)] for real-time SHM do not report their updating frequency. As a result, the real-time characteristic of those systems remains unknown, resulting in (1) a low trustworthiness of those systems for practical implementation in SHM, and (2) low applicability because unknown updating frequency cannot properly and optimally inform the scheduling of structural maintenance and rehabilitation activities. The larger the updating frequency (shown in Table 3) the better since higher frequency reflects that the data is collected, processed, and visualized at a shorted period of time (i.e., more frequently). Although a few studies presented the updating frequency for their real-time SHM systems, the updating frequency values are still small, which means observable lags still exist in those systems. For example, the system by Wang et al. (2022) has a high latency due to an updating frequency of each day (1.15×10^{-5} Hz). This study specifically reaches a high updating frequency of once per 0.2 s (i.e., 5 Hz) to complete an entire process of sensing and preprocessing the sensor data, handling data requests, processing the data, and updating the digital model in BIM to visualize the structural behaviors, which is considerably higher from existing reported frequencies in the literature.

From a practical implementation perspective, this high digital twin updating frequency of 5 Hz can enhance the effectiveness of real-time monitoring and decision making in two main ways. First, it provides real-time insights into the structural assessment, offering both engineers and stakeholders access to the current structural conditions with a delay as small as 0.2 s. As a result, more efficient maintenance decision can be made to save time by avoiding prolonged deliberations and indecision, optimize the allocation of resources (e.g., materials, and manpower), and improve productivity of structural maintenance-related activities. Second, capitalizing on the high updating frequency, it enables early detection of any subtle changes or anomalies in the structural behaviors, thus further reducing the risks of a structural health issue escalating from minor to major problems. Therefore, the proposed digital twin-based SHM framework has not only the technical advancement based on a higher updating frequency but also greater impact on the practicality of digital twins for SHM.

5.2. Contributions

This paper contributes to the body of knowledge in three main ways.

First, it proposes a BIM-enabled digital twin framework with much higher updating frequency than the existing similar BIM-based real-time SHM systems, advancing the development of digital twins-based SHM toward faster and more adaptive direction. The proposed framework can retrieve, analyze, and visualize the real-time data on the performance

Table 3
Most recent studies for developing digital twin-based and/or real-time SHM systems.

Reference	Digital twins	Application	Sensor data type	Updating frequency
Rojas-Mercedes et al. (2022)	Y[a]	Seismic assessment of concrete bridge SHM	Triaxial acceleration signals	NA*
Dang et al. (2022)	Y	Steel truss bridge SHM	Triaxial acceleration signals	NA
Wang et al. (2022)	N[b]	Timber heritage structure SHM	Fiber Bragg grating strain measurements	(1.15×10^{-5} Hz) 1 day
Fawad et al. (2023)	N	Bridge SHM	Vibrating wire strain gauge signals	NA
Teng et al. (2023)	Y	Steel truss bridge damage detection	Triaxial acceleration signals	NA
Hielscher et al. (2023)	Y	Concrete footbridge SHM	Fiber optical sensor-based strain measurements	NA
Longman et al. (2023)	Y	Timber shear wall tension loss and moisture content monitoring	Hydrothermal signals and moisture measurements	NA
Sakr and Sadhu (2023)	Y	Bridge and building SHM	Triaxial acceleration signals	1/35 Hz
This study	Y	Structural tension strain and deflection monitoring	Strain gauge-based strain measurements and laser distance signals	5 Hz

[a] Y: a digital twin-based SHM system;
[b] N: not a digital twin-based but a real-time SHM system;
* NA: the information is not available or applicable.

and behavior of physical assets or systems in a faster manner, allowing for immediate detection of anomalies, optimization opportunities, and preventive maintenance. Meanwhile, this framework can also be used for monitoring the health conditions of structural components (e.g., a bridge girder with built-in redundancy and safety margins) that do not require fast responses, thus showing its high adaptability to different application scenarios.

Second, this study incorporates digital twins with BIM interoperability that allows for seamless real-time SHM data exchange for supporting structural analysis, smart sensing, and structural behavior visualization. The proposed approach requires the development of a digital model of the physical structure only once, which can then be used for both structural analysis (to identify optimal sensor location) and structural behavior visualization in another BIM software when sensor data streams in dynamically. Such BIM model interoperability ultimately ensures that all stakeholders can collaborate effectively during the entire lifecycle of a monitored structure and that the data across different BIM software remains with high consistency and accuracy.

Finally, this study innovatively integrates the rich structural analysis-related domain knowledge with sensing data analytics to unleash the potential of existing BIM software for more intuitive structural behavior representation and visualization. This integration specifically leverages domain knowledge to perform sensor data interpolation, thus increasing the cost-effectiveness of the IoT-based SHM systems. It specifically minimizes the use of sensors while maintaining the high representativeness of the structural behavior visualization by simultaneously capturing different behaviors at different locations of a structure.

5.3. Limitations

Three limitations are acknowledged in this study. First, the proposed structural analysis domain knowledge-based strain sensor data interpolation method has limitation on interpolating the data for a structural element when live load with varying locations are applied. Given the information of the loading conditions and boundary conditions, it is straightforward to formulate the mathematical representations [i.e., Eq. (6) and Eq. (10) for enabling automated sensor data interpolation]. In fact, the mathematical representations need to be pre-defined by system developers based on structural analysis domain knowledge. However, when live loads with changing locations occur, it will result in the changes in the mathematical representations, leading to a challenge of dynamically defining the mathematical representations by precisely locating the live loads. Thus, it is impractical to update those mathematical representations to fit the scenarios in which the live loads have changing locations. Then, it exposes another research need to address the strain sensor data interpolation challenge caused by the varying locations of live loads. Second, since the scope of this study is to perform tensile strain sensor data interpolation and visualization, it currently does not consider other structural behaviors such as shear strain induced by torsional loads. As a result, the monitored health conditions of a structural element by the proposed SHM system may not comprehensively reflect the overall health conditions of the structural elements. It further points out that future work can focus on expanding the smart sensing unit by adding more sensors to sense heterogenous structural behaviors under more complex loading conditions. Third, the proposed BIM-enabled digital framework system was tested on a small-scale structural frame in a controlled environment. The applicability of the proposed system for real-time scenarios still remains uncertain. To ensure the applicability of the proposed system for real-world structures, in the future, the authors will conduct additional validations of the system's applicability through full-scale structural test in the laboratory or in the field by considering more types of structural geometries and material properties.

6. Conclusions

This study proposed an innovative BIM-enabled digital twin framework that embeds IoT, wireless communication, digital signal processing, and structural analysis domain knowledge for SHM. The proposed framework was tested by monitoring and visualizing the deformation of critical structural elements (i.e., beam, and column) of a prototyped structural frame using strain gauge and laser displacement sensor measurements. Different load configurations are applied on the frame protype to validate the proposed framework. The results showed that the developed digital model of the structural frame prototype has high interoperability as it can be used across different BIM software tools (i.e., Robot Structural Analysis Professional and Revit). This model was used for (1) performing structural analysis to identify the most vulnerable locations on a structure in order to inform optimal locations for sensor installation, and (2) guiding the creating of a DT model for real-time structural behavior visualization. By comparing two different digital signal processing techniques (FFT, and WT), this study found that WT with 'sym4' kernel (WCT = 25; DL = 6) was the optimal technique to handle the noises in the laser sensor signals. Thus, it further proves the significance of signal preprocessing to address the potential sensor data quality issues in IoT-based SHM. With the use of visual programming, BIM, and structural analysis domain knowledge, this paper creatively proposed a sensor data interpolation method in Revit to visualize the diverse structural behaviors of a structural element using measurements from one single sensor only. In Revit, the real-time visualization of the beam deformation and column displacement under different loads was smooth as the digital twin model could refresh at a frequency of 5 Hz. Meanwhile, the knowledge-based sensor interpolation enabled an intuitive beam deformation visualization as the deformations at different

locations of a beam were visualized differently using a color-coding scheme. This paper ultimately contributes to the body of knowledge by developing a structural analysis expertise-guided BIM-enabled digital twin framework for SHM with reliable sensor data quality and intuitive visualization capability to enhance the existing practices of BIM-based SHM for civil built asset management.

CRediT authorship contribution statement

Xi Hu: Conceptualization, Data curation, Methodology, Formal analysis, Investigation, Writing – original draft, Visualization. **Gulsah Olgun:** Conceptualization, Data curation, Methodology, Formal analysis, Investigation, Writing – original draft. **Rayan H. Assaad:** Conceptualization, Methodology, Formal analysis, Writing – review & editing, Project administration, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The paper contains all the details related to the data, models, and programming scripts used in this research.

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